Comparison of straining during defecation in three positions: results and implications for human health.

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The aim of the study was to compare the straining forces applied when sitting or squatting during defecation. Twenty-eight apparently healthy volunteers (ages 17-66 years) with normal bowel function were asked to use a digital timer to record the net time needed for sensation of satisfactory emptying while defecating in three alternative positions: sitting on a standard-sized toilet seat (41-42 cm high), sitting on a lower toilet seat (31-32 cm high), and squatting. They were also asked to note their subjective impression of the intensity of the defecation effort. Six consecutive bowel movements were recorded in each position. Both the time needed for sensation of satisfactory bowel emptying and the degree of subjectively assessed straining in the squatting position were reduced sharply in all volunteers compared with both sitting positions ($P < 0.0001$). In conclusion, the present study confirmed that sensation of satisfactory bowel emptying in sitting defecation posture necessitates excessive expulsive effort compared to the squatting posture.